



AI SOFTWARE DEVELOPMENT INTERNSHIP

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Week 6 — Detailed Task Sheet

LangChain & LCEL • Advanced Retrieval • Tool Calling Resilient .NET fi AI
Service Integration • End-to-End Streaming • Git Rebase

PART D — TOOL CALLING WITH LANGCHAIN

Outputs

```
• (venv) PS D:\Week6\Week6_PartD_Tool_Calling> python tool_calling.py

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PART D - LANGCHAIN TOOL CALLING
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QUESTION:
Is book 8 available right now?
=====
D:\Week6\Week6_PartD_Tool_Calling\venv\Lib\site-packages\langchain_google_genai\chat_models.py:3908: UserWarning: Model 'gemini-3.5-flash-lite' uses fixed
sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(

TOOL CALL DETECTED:
Tool name: check_book_availability
Arguments: {'book_id': 8}
Tool result: Currently borrowed

=====
QUESTION:
What genre is book 8?
=====
D:\Week6\Week6_PartD_Tool_Calling\venv\Lib\site-packages\langchain_google_genai\chat_models.py:3908: UserWarning: Model 'gemini-3.5-flash-lite' uses fixed
sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(

NO TOOL CALL
Model response:
[{'type': 'text', 'text': "I cannot answer your question directly about the genre of book 8, as I don't have access to that information in my current datab
ase, and I am only permitted to check whether books are available to borrow.", 'extras': {'signature': 'EmAKXgFpFH0T51oIkSozh8s4xDuj8rxyh26CLG8sAVRi9QI/1BH
ZlqXtJmHRj8K9F+6wFnYB0X4vrdFSp+OyrAZfDhQI7bTSrw2fHTCUQJHqXwhlmpNILU8G+Ww40Vphs='}}]
○ (venv) PS D:\Week6\Week6_PartD_Tool_Calling>
```

```
• (venv) PS D:\Week6\Week6_PartD_Tool_Calling> python tool_calling.py

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PART D - TOOL RESULT FED BACK TO MODEL
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USER QUESTION:
Is book 8 available right now?
=====
D:\Week6\Week6_PartD_Tool_Calling\venv\Lib\site-packages\langchain_google_genai\chat_models.py:3908: UserWarning: Model 'gemini-3.5-flash-lite' uses
sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(

MODEL TOOL DECISION:
Tool name: check_book_availability
Arguments: {'book_id': 8}
Tool result: Currently borrowed
D:\Week6\Week6_PartD_Tool_Calling\venv\Lib\site-packages\langchain_google_genai\chat_models.py:3908: UserWarning: Model 'gemini-3.5-flash-lite' uses
sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(

FINAL NATURAL-LANGUAGE ANSWER:
No, book 8 is currently borrowed and not available right now.
```